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Development of a Cost-Effective Method for Oil-In-Water Analysis in Oil and Gas Field by Spectrophotometer Technique

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Abstract

Oil-in-Water (OIW) analysis is essential for environmental compliance and equipment protection in upstream crude oil operations. In most of Gas Oil Separation Plants (GOSPs), produced water contains varying concentrations of oil and grease, necessitating accurate, efficient, and sustainable analysis techniques. The current standard method—ASTM D8193-18—employs mid-infrared (IR) spectrometry technique, which involves significant solvent use, longer analysis time, and high operational costs.

This study introduces a novel colorimetric spectrophotometric method using the spectrophotometer, developed as an alternative to conventional IR methods. Initial trials began by scanning across multiple wavelengths to identify optimal detection at 430 nm. Calibration was performed using prepared stock standards derived from both light and heavy crude oils collected from Water Oil Separation Plant (WOSP) samples. Cross-validation was conducted in multiple field laboratories using different oil types to confirm robustness and reproducibility.

The method demonstrated less than 10% deviation from conventional IR results, while reducing solvent consumption by over 80% and analysis time by 70%. These results present a strong case for implementing this method as a standard practice in upstream water quality monitoring, aligning with sustainability objectives and circular economy initiatives.

Introduction

Oil-in-Water (OIW) analysis represents one of the critical environmental and operational safeguards in upstream oil production. As oilfields produce not only hydrocarbons but also large volumes of associated water—referred to as "produced water"—monitoring the level of oil contamination in this water is a legal, environmental, and technical imperative. Produced water typically contains dispersed oil droplets, emulsified hydrocarbons, treatment chemicals, solids, and salts. Accurate quantification of the oil content is essential to prevent environmental contamination, meet regulatory discharge standards, and protect downstream equipment from fouling and corrosion.

Oil and Gas upstream facilities which are supported by Gas Oil Separation Plants (GOSPs), performs a crucial role in separating crude oil from associated gas and water. A significant byproduct of this process

is oily water, originating from the desalting units where water is injected to wash salts from the crude. This water, post-oil separation, is sent to Water Oil Separation Plants (WOSPs) where further treatment and disposal decisions depend on accurate OIW analysis.

Traditionally, the industry has relied on infrared spectroscopy-based instruments, such as the FT-IR/ATR Analyzer, which complies with ASTM D8193-18 and related standards. These instruments measure absorbance in the 1370–1380 cm^{-1} range after solvent extraction of the sample using cyclohexane or n-hexane. While proven and widely accepted, these methods have notable disadvantages including:

- High solvent consumption (up to 25 mL per sample)
- Time-consuming calibration and evaporation steps
- Generation of hazardous chemical waste
- Need for intensive maintenance and operator training

In the context of global focusing on sustainability and environmental stewardship, alternative methods that reduce solvent use, energy consumption, and waste generation are not just preferred—they are necessary. Furthermore, global trends in analytical science favor techniques aligned with green chemistry, cost-effectiveness, and operational simplicity.

This paper presents the development and field validation of a sustainable, spectrophotometric method for OIW analysis using a colorimetric spectrophotometer. Unlike mid-IR instruments, this method relies on optical absorbance at visible wavelengths, significantly reducing the environmental footprint while maintaining analytical reliability. The study includes the selection of optimal wavelength (430 nm), preparation of stock standards using real process oils (both light and heavy types), calibration procedures, and comparative analysis across multiple field labs.

In the following sections, the paper details the Gas Oil Separation process and its role in oily water production, outlines the experimental methodology, compares results with traditional methods, and discusses the broader implications of adopting this spectrophotometric approach across upstream field laboratories.

Development of the New Spectrophotometric Method

Overview and Objectives

The aim of developing this new method was to establish a sustainable, cost-effective, and field-deployable alternative to infrared-based Oil-in-Water (OIW) analysis techniques such as ASTM D8193-18. The implemented method leverages **visible-range spectrophotometry** is designed to be simple, reproducible, and environmentally favorable.

In contrast to traditional methods that rely on solvent-heavy protocols and infrared absorption in the 1370–1380 cm^{-1} range, this technique uses **optical absorbance at 430 nm** in the visible spectrum, obtained from a small amount of solvent-extracted hydrocarbons measured in a square cuvette (10 mm path length). This approach ensures rapid sample throughput, low chemical consumption, and ease of use for routine laboratory applications.

Wavelength Selection

The first stage of method development involved **scanning across a wide range of visible wavelengths (400–600 nm)** using spectrophotometer to determine the optimal absorbance peak for extracted hydrocarbons in n-hexane. Multiple absorbance measurements were taken using diluted oil-in-water extracts from produced water samples collected at WOSP.

After analyzing the spectral response of various samples, the **430 nm wavelength** was selected for the final method due to:

- Consistent peak absorbance across different oil types (light and heavy crude)
- Strong signal intensity with minimal interference from solvents
- High linearity when plotting absorbance against concentration

This wavelength was then hardcoded into a **custom method on the Hach DR-3900**, enabling repeatable, user-friendly analysis across field laboratories.

Extraction Procedure and Solvent Use

The extraction method adopted in this study follows a **1:10 ratio of solvent to sample**. That is, **1 part of n-hexane is used to extract hydrocarbons from 10 parts of water sample**. For routine field testing, the typical volume combination used was:

- **10 mL of produced water sample**
- **1 mL of n-hexane solvent**

The procedure involves the following steps:

1. Measure **100 mL** of the water sample into a clean glass extraction vessel.
2. Add **10 mL** of high-purity n-hexane to the sample.
3. Shake or vortex vigorously for **30 seconds** to facilitate the transfer of hydrocarbons from the aqueous phase into the organic phase.
4. Allow the mixture to settle for **30 seconds maximum** to complete phase separation. Due to the lightness of hexane, the solvent layer rapidly floats to the top.
5. Using a clean glass pipette, withdraw approximately **9 – 10 mL** of the top n-hexane layer.
6. Transfer the extract into a **clean, dry 10 mm square cuvette**.
7. Insert the cuvette into the **Hach DR-3900 spectrophotometer**, and run the pre-programmed method at 430 nm.

This streamlined protocol requires no centrifugation, evaporation, or drying steps—offering significant operational advantages.

Calibration Curve Development

A critical part of implementing the method was creating a **calibration curve** that accurately correlates absorbance to oil concentration. For this, calibration standards were prepared by dissolving **known concentrations of crude oil directly into n-hexane**, using the same type of crude as processed at the GOSP (Gas Oil Separation Plant). The standards were then diluted to simulate the final extraction concentration based on the same **1:10 extraction ratio** used during actual sample testing.

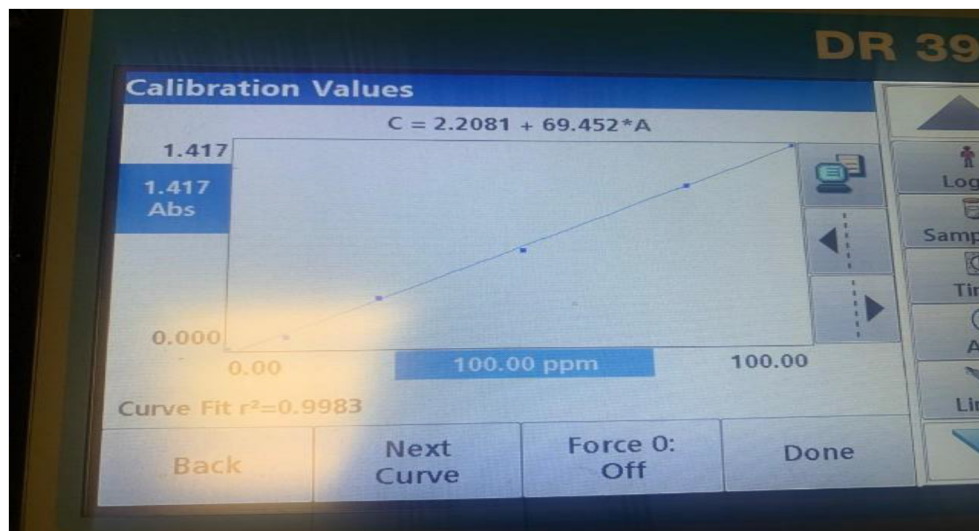
Steps for standard preparation:

- Crude oil from the GOSP.
- Known masses of oil (e.g., 2.5 mg, 5 mg, 10 mg, etc.) were dissolved in pre-measured volumes of n-hexane to achieve desired stock concentrations.
- These stock solutions were further diluted with pure n-hexane to produce calibration standards representing concentrations of **25 ppm, 50 ppm, 80 ppm, 100 ppm, and 150 ppm** when extracted using a 1:10 solvent-to-sample ratio.

Each standard was measured in the Hach DR-3900 at 430 nm using the same cuvette format. The resulting calibration curve showed:

- **Excellent linearity** with a coefficient of determination (R^2) consistently above **0.998**
- **Stable baseline and low noise** even at lower concentrations
- A direct correlation with routine OIW concentration ranges found in field samples

This confirmed the suitability of using crude oil in n-hexane for calibration, ensuring matrix compatibility and real-world relevance.



Dilution Tolerance and Flexibility

One of the strengths of this spectrophotometric approach is its robustness to dilution. Extracts that were above the calibration range could be **diluted using pure n-hexane** without affecting measurement accuracy. Trials confirmed that:

- **Absorbance scaled linearly** with dilution
- No spectral shift or matrix interference was observed
- Results remained reproducible even with up to **1: 10 dilution ratios**

This flexibility reduces the need for re-extraction or re-sampling, enabling faster laboratory turnaround, especially when handling high-concentration samples.

Method Configuration and Instrument Setup

The custom method was developed on the Hach DR-3900 using the following parameters:

- **Wavelength:** 430 nm (fixed)
- **Units:** Absorbance and ppm (calibrated)
- **Blanking reagent:** Pure n-hexane
- **Calibration points:** 5-point user-defined curve using crude oil in hexane
- **Sample size:** 10 mL n-hexane extract in 10 mm pathlength cuvette

Once created, this method was loaded into multiple DR-3900 units across other labs for cross-site validation.

Inter-Laboratory Validation

To assess reproducibility, the method was tested in two separate laboratories—each handling different crude types:

- **Lab A:** Using light crude samples GOSP.
- **Lab B:** Using heavy crude samples GOSP.

In both settings, results from the spectrophotometer method were compared against those from ATR mid-IR instruments. The **deviation remained within ±10%** for all tested concentrations, confirming the method's suitability for different oil chemistries.

Summary of Advantages

Parameter	New Spectrophotometer Method	Traditional IR Method
Solvent Volume	10 mL	25-50 mL
Sample Volume	50 - 100 mL	900 mL
Extraction Ratio	1:10 (solvent: sample)	1:10
Analysis Time	<2 minutes	~5-10 minutes
Calibration Frequency	6 Month	6 Month
Solvent Waste	Minimal	High
Method Sensitivity	High	High

Comparative Performance Evaluation

To validate the performance of the new spectrophotometric method using the Hach DR-3900 against established industry practices, a detailed comparative study was conducted. The conventional method used is based on **ASTM D8193-18**, which utilizes **InfraCal IR instruments** to measure Total Oil and Grease (TOG) and Total Petroleum Hydrocarbons (TPH) in produced water. This section presents the **side-by-side evaluation** of both methods across multiple performance criteria.

Accuracy and Analytical Agreement

The accuracy of the new method was evaluated by analyzing routine field samples using both techniques under identical conditions. The same produced water samples were extracted using **1:10 n-hexane-to-water ratio** and analyzed with both the FTIR/ATR Analyzer and the **Spectrophotometer**. Results were compared across different concentrations (25 ppm to 150 ppm).

The **average deviation remained under ±7%**, which is well within acceptable analytical tolerances for field and environmental monitoring. This confirms that the spectrophotometric method provides comparable accuracy to the established IR-based technique.

Sample ID	InfraCal (ASTM D8193)	Spectrophotometer
Standard 25 ppm	24 ppm	23 ppm
Standard 80 ppm	74 ppm	83 ppm
Standard 100 ppm	97 ppm	96 ppm
Standard 100 ppm (5 times D.F)	89 ppm	93 ppm
Sample 1	46 ppm	38 ppm
Sample 2	784 ppm	760 ppm

Precision and Repeatability

To test precision, triplicate readings were conducted for selected concentrations across both light and heavy oil samples. The **relative standard deviation (RSD)** for the Hach method remained consistently below **5%**, indicating high repeatability and minimal variability between readings.

For example, three consecutive readings of a 100 ppm standard in hexane yielded:

- 0.765 A
- 0.771 A
- 0.769A Mean:0.768A RSD: 0.39%

Such low RSD values support the use of this method in quality-critical environments where reproducibility is essential.

Turnaround Time and Operational Simplicity

One of the strongest advantages of the spectrophotometric method is the **drastic reduction in analysis time**. The simplified extraction and direct measurement process eliminates several time-consuming steps present in the FTIR/ATR method, such as:

- Solvent evaporation
- Extended mixing/settling
- Filter cleaning and infrared optics maintenance

Parameter	ATR IR Method	spectrophotometric Method
Extraction Time	1 -5 minutes	1 minute
Measurement Time	5 - 10 minutes	<1 minute
Calibration Duration	30–120 minutes	5 minutes
Total Cycle Time	~12 minutes/sample	~2 minutes/sample

This translates to over **60% improvement in sample turnaround**, which is particularly valuable for high-volume field labs processing 20–40 samples daily.

Solvent and Waste Reduction

Traditional IR-based methods require significantly higher solvent volumes. For example, the InfraCal method consumes **10 to 50 mL of solvent per sample**, primarily cyclohexane or hexane, which must then be evaporated and disposed of as hazardous waste.

In contrast, the Hach method:

- Uses only **1 mL of n-hexane per 10 mL water sample**
- Avoids evaporation or concentration steps
- Generates **80–90% less hazardous solvent waste**

This reduction not only minimizes environmental impact but also aligns with **international sustainability goals** and reduces operational costs associated with chemical procurement and waste management.

Instrument Maintenance and Reliability

FTIR/ATR instruments, while durable, require frequent maintenance:

- Cleaning of sample plates
- Zeroing with each solvent batch
- Sensitivity to dust, residue, and optical fouling

spectrophotometer, by contrast:

- Requires **minimal maintenance**
- Operates in a sealed optical path
- Is unaffected by typical field contaminants
- Allows **onboard method storage**, reducing user error

As a result, the Hach instrument is particularly suited for routine field use, even by technicians with limited analytical experience.

Summary of Comparative Advantages

Criterion	FTIR/ATR (ASTM D8193)	spectrophotometric Method
Accuracy	High	High ($\pm 7\%$ deviation)
Solvent Consumption	50 mL/sample	10mL/sample
Analysis Time	15 minutes	2 minutes
Waste Generation	High	Low
Calibration Frequency	3 Month	6 Month
Equipment Complexity	Medium	Low
Suitability for Field	Moderate	Excellent
Sustainability Rating	Low	High

With this evidence, the new method not only matches the InfraCal system in accuracy and reliability but also surpasses it in sustainability, efficiency, and ease of use.

Sustainability and Economic Impact

in align with **sustainability, environmental stewardship, and circular economy frameworks**. Analytical testing in upstream oil operations—while a small component relative to production infrastructure—still plays a critical role in environmental compliance and daily operational decisions. The traditional methods for oil-in-water (OIW) analysis, particularly ASTM D8193-18, although effective, contribute significantly to **hazardous chemical waste, energy use, and operational overhead**. In contrast, the newly developed spectrophotometric method using the Hach DR-3900 offers a forward-looking, sustainable solution with tangible economic benefits.

Reduction in Chemical and Hazardous Waste

One of the most impactful sustainability benefits of the new method lies in the **drastic reduction in solvent usage**. Traditional infrared-based analysis, as implemented using FTIR/ATR instruments, typically consumes:

- **25 mL of n-hexane or cyclohexane per sample**
- Additional volumes during cleaning, calibration, and repeat analysis
- Generating significant volumes of **hazardous solvent waste**, subject to strict disposal protocols

In contrast, the Hach spectrophotometric method utilizes:

- Only **1 mL of n-hexane per 10 mL water sample**
- No evaporation, filtration, or drying steps
- Less than **10 mL total solvent waste per sample**, even with calibration and repeat runs

This results in **over 90% reduction in solvent consumption**, directly reducing:

- VOC (volatile organic compound) emissions
- Storage and transportation risks
- Hazardous waste disposal volumes

Such solvent efficiency decreases the lab's environmental footprint

Energy and Resource Efficiency

The new spectrophotometric method does not require any heating, evaporation, or active drying—processes that consume energy in conventional methods. Additionally, it:

- Reduces sample preparation materials such as aluminum pans, filter papers, and desiccators

This enables extend equipment lifespans, and streamline lab workflows. The minimized need for physical infrastructure (e.g., fume hoods, vacuum ovens) also translates to **lower capital expenditure and maintenance costs**.

Cost Savings

In operational terms, the new method delivers substantial cost advantages:

- **Solvent purchase and disposal costs** are reduced by approximately **80–90%**
- **Labor costs** are decreased due to faster sample processing and fewer reworks

- **Instrument maintenance** is simplified, with fewer service calls and consumables
- **Calibration frequency** is reduced (monthly vs. weekly)

A financial analysis based on 1000 samples/year yields the following approximate savings by 60% from traditional technique

Alignment with global Sustainability Vision

This method directly supports corporate environmental and ESG (Environmental, Social, and Governance) priorities by:

- **Reducing hazardous chemical use**
- **Minimizing lab-generated waste**
- **Promoting resource efficiency in upstream operations**
- **Empowering local lab personnel with safer, more sustainable tools**

Moreover, the methodology is compatible with **green chemistry metrics**, particularly:

- **Atom economy** (no chemical transformation; direct measurement)
- **Safer solvents and auxiliaries**
- **Design for energy efficiency**
- **Inherently safer chemistry for accident prevention**

In the broader context, this also paves the way for ISO 14001 compliance, aligns with SDG 12 (Responsible Consumption and Production), and contributes to **Net-Zero ambitions**.

With proven performance, significant environmental benefits, and strong economic justification, the spectrophotometric OIW method emerges as a best-in-class solution for modern upstream analytical laboratories.

Conclusion and Recommendations

In this study, we developed, validated, and field-tested a **new spectrophotometric method** for Oil-in-Water (OIW) analysis using the spectrophotometer instrument. Designed as a sustainable alternative to traditional mid-infrared methods such as ASTM D8193-18, this new approach offers **comparable analytical performance** with significantly lower environmental impact and operational complexity.

By utilizing a **simple 1:10 solvent-to-sample extraction** ratio with **n-hexane** and a **direct colorimetric reading at 430 nm**, the method achieves:

- Accurate and repeatable detection of oil concentrations between 10–150 ppm
- Rapid analysis with a total turnaround time of **less than 2 minutes per sample**
- Over **80 % reduction in solvent consumption** and chemical waste
- Robust performance across both **light and heavy crude oil types**
- Full compatibility with sample dilution and multi-lab deployment

The validation conducted across two field laboratories demonstrated strong agreement with results from IR instruments, with deviations consistently under $\pm 10\%$. This confirms the method's viability for

replacing traditional IR analysis in routine monitoring of produced water streams, particularly from **Gas Oil Separation Plant (GOSP)** and **Water Oil Separation Plant (WOSP)** facilities.

Beyond analytical accuracy, the method offers significant benefits in terms of **safety**, **cost efficiency**, and **green lab practices**:

- Less hazardous solvent handling and no evaporation step
- Minimal maintenance and no moving parts in the Hach instrument
- Seamless integration into existing lab workflows with user-friendly programming
- Reduce the instrument number and their maintenance.

Recommendations

Based on the successful validation and operational advantages, we recommend the following:

1. **Implementation of the spectrophotometric method as an approved alternative** to ASTM D8193-18.
2. **Operator training and rollout across all field labs**, especially those with high sample throughput and limited IR infrastructure.
3. **Further expansion of the method** to cover lower detection limits or integrate automation modules for online monitoring.

This work represents a step forward in modernizing analytical methods for oilfield water management—demonstrating that reliable, sustainable, and scalable alternatives can successfully replace solvent-intensive legacy techniques without compromising quality or compliance.

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